

GUFAN WANG

Robotics Engineering B.Eng. Candidate
Legged Locomotion | Sim-to-Real | Real-Time Robotic Systems

gufan.wang.ustb@gmail.com
robotics-research-portfolio.vercel.app
ORCID: 0009-0006-8411-388X

Education

University of Science and Technology Beijing (USTB)

B.Eng. in Robotics Engineering; GPA: 3.5/4.0

Beijing, China

Sep 2023 - Jun 2027

Relevant coursework: Robotics, Automatic Control Theory, Machine Learning, Reinforcement Learning and Optimal Control, Deep Learning, Mechanical Design.

Robotics Research and Engineering Experience

25 kg Wheel-Legged Robot - RL Locomotion and Real-Robot Deployment

May 2026 - Aug 2026

USTB Advanced Robotics Laboratory

- Built the real-time deployment stack for a 25 kg wheel-legged robot, connecting a PPO policy to 16 actuators through Linux shared memory, two STM32F407 controllers, and four CAN buses.
- Trained and deployed standing, wheeled locomotion, stair traversal, and disturbance-recovery policies in Isaac Lab using a 57-D observation and 16-D action interface shared by simulation and hardware.
- Resolved disturbance-induced oscillation on the physical robot by broadening dynamics randomization and retuning joint stiffness and damping; the deployment loop enforced synchronization, feedback-freshness, and watchdog safeguards.

55 kg Quadruped - PPO Locomotion and EtherCAT Sim-to-Real

Sep 2025 - Jun 2026

USTB Advanced Robotics Laboratory; in collaboration with researchers from Tsinghua University

- Deployed a PPO locomotion controller on a 55 kg quadruped, with policy inference at 50 Hz, joint PD control at 1 kHz, and a SOEM-based EtherCAT master exchanging PDOs at 2 kHz.
- Integrated 12 CiA 402 drives in Profile Torque Mode, IMU and joint feedback, action mapping, state-machine logic, and shared-memory interfaces into the hardware control stack.
- Built joint-command and feedback logging to isolate model error, actuator delay, static friction, and interface mismatches during sim-to-real debugging.

20-Motor Upper-Limb Exoskeleton - EtherCAT Control-System Migration

Sep 2024 - Jan 2025

USTB Advanced Robotics Laboratory

- Migrated a 20-motor real-time control system from Windows to Ubuntu and ROS 2, reworking EtherCAT communication and motor-control interfaces.
- Integrated a control card without native Linux support and connected data-acquisition and command paths to the ROS 2 control stack.

Field Robotics Experience

Magnetic Wall-Climbing Inspection Robot - Vision and Field Deployment

Sep 2024 - Sep 2025

USTB Advanced Robotics Laboratory; Wudongde Hydropower Station field deployment

- Contributed to metal-surface defect data collection, model comparison, and YOLO11 deployment for a magnetic climbing inspection robot.
- Participated in field deployment under dust, vibration, illumination variation, and viewpoint constraints.

Scientific Machine Learning Experience

Time-Varying PINN for Furnace Temperature Prediction

Jun 2025 - Dec 2025

USTB Automation and Control Laboratory

- Extended a static PINN workflow to full-cycle temperature-field prediction under nonuniform heat-source conditions; combined a 2D finite-difference solver with PDE and boundary-condition losses.

Data-Driven Magnesium Alloy Property Modeling

Jun 2025 - Sep 2025

Taiyuan University of Technology Forming Laboratory

- Contributed to data-augmentation and regression workflows using Transformer, MLP, and stacking models for alloy-property prediction from composition and process variables.

Research Outputs

- PINN-Based Material Temperature Forecasting and Heating-Furnace Curve Optimization - second author; accepted for publication in Measurement, 2026.
- Efficient Design of Magnesium Alloys with Superior Mechanical Properties Using Deep Learning - second author; manuscript under review.
- Two Chinese utility model patents - one as sole inventor and one as co-inventor.

Technical Skills

Programming and real-time systems: C/C++, Python; Linux, ROS 2, shared-memory IPC; STM32, CAN, and EtherCAT integration.

Reinforcement learning and simulation: PyTorch, PPO, Isaac Lab, and MuJoCo; sim-to-real transfer and dynamics randomization.

Scientific computing and perception: PINNs, finite-difference modeling, YOLO-based detection.

Honors and Awards

- National First Prize, China International College Innovation Competition, 2025.
- Beijing Grand Prize, Challenge Cup AI Track, 2025; National Third Prize, Challenge Cup, 2025.
- USTB Freshman Scholarship and People's Scholarship (Third Class).